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Network Design for NorahMedical Clinic

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Technologies

Case Study	
Communication Networks (ECE 380)	
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Executive Summary

This project explains the design and implementation of a complete communication network for NorahMedical Clinic using Cisco Packet Tracer. There are two floors in the clinic that have multiple departments (Administration, Reception, Doctor's Workstations, Medical lab). We developed our network to be well-organized, expandable, safe and easily managed while being affordable on hardware costs. We utilized VLAN (Virtual Local Area Network) technology to divide departments based on departmental needs. When there are multiple departments connected to the same physical switch, they can be divided logically using VLAN technology. We created sub-interfaces on the router interfaces and trunk ports on the switches' ports to allow each of the VLANs to communicate with each other through one connection per floor as opposed to having one cable for every VLAN. This simplified cabling complexity and hardware usage but still maintained organization and separation throughout the network. Additionally, this project includes many of the advanced concepts beyond just what is required to meet the basic objectives such as remote home access via a VPN (Virtual Private Network), separating VLANs within the home network, and ACLs (access control lists). By implementing a VPN (Virtual Private Network) it completely secured routing from the clinic network to the manager's home network as it encrypted all routed packets.

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1. Introduction

In this project we created an entire communication system in a fully functional version at Norah Medical Clinic utilizing Cisco Packet Tracer. There are two floors in the building with many departmental locations which require communication between each location on their respective floors but also need to be completely separated so as to provide the best organizational and security features possible. Our network is comprised of many different components such as routers, switches, VLANs, trunk connections and sub-interfaces. Our goal of the project was to create a network that would be secure yet well-organized and easy to understand and maintain while being economically feasible. First, we completed a case study of our client's current state of affairs (NorahMedical Clinic). At that time, they did not have a functioning network, so all aspects of their new network had to be configured by us. These include designing the topology of the network, the VLAN structure for separation of devices into logical groups, how to sub-net the available IP addresses, the addressing plan for all of the IP enabled devices within the network and finally the configuration of all of the individual devices.

2. Problem Statement

NorahMedical Clinic recently moved into a new building and needed a completely new network infrastructure. The manager wanted all departments connected together while still keeping each department separated from the others. The clinic contains administration offices, reception area, doctors workstations, and medical laboratory distributed across two floors. Every department needed its own subnet and VLAN while still being able to communicate through the router. The manager also wanted the switches to be remotely manageable and required all devices to communicate successfully after the network configuration was completed. Future growth was also important because the clinic may expand later by adding more departments or devices. Another important requirement was keeping the design simple and cost efficient. Because of this, the network had to be designed carefully to reduce unnecessary hardware and cables while still keeping good performance and reliability.

3. Network Requirements Analysis

3.1 Functional Requirements

The network needed to support communication between all devices and departments inside the clinic. Every department needed its own VLAN and subnet while still being able to communicate through the router. The switches also needed management IP addresses so administrators can manage them remotely. Printers needed to work correctly inside their assigned departments and all ping tests needed to succeed between devices. The network also needed to support communication between the clinic network and the manager's home network.

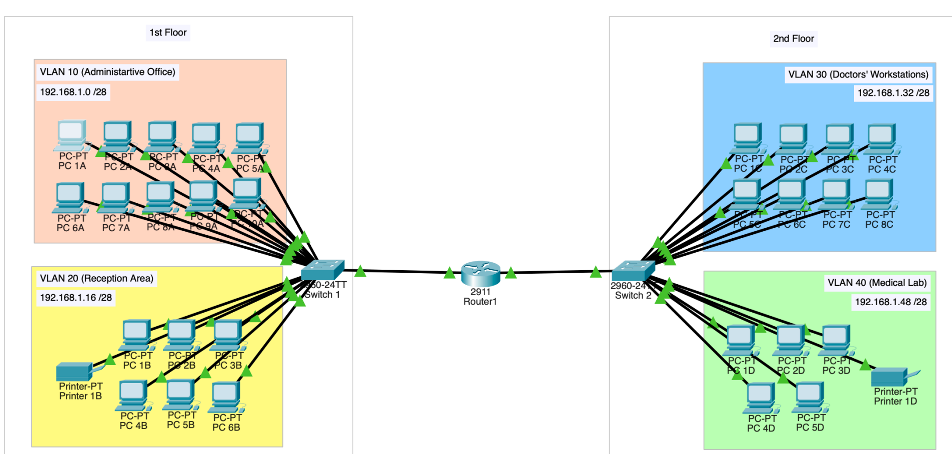
3.2 Non-Functional Requirements

Besides the basic functionality, the network also needed to satisfy several non-functional requirements. The design needed to remain simple and organized because the clinic is relatively small and does not require complicated enterprise hardware. At the same time, the network needed to support future expansion without needing complete redesign later. Security was another important requirement. Departments needed to remain separated logically and home devices should not access company resources. The network also needed to reduce unnecessary traffic and improve troubleshooting. Reliability was also considered during the design process. This is one of the reasons why two separate physical connections were used between the router and the switches instead of depending on one single connection.

4. Proposed Network Design

4.1 Selected Topology

The final topology contains one router, two switches, 29 PCs, two printers, and a separate home network for the clinic manager. Switch 1A was assigned to Floor 1 while Switch 2A was assigned to Floor 2. Even though only two switches were used, the clinic was still divided into four separate departments using VLAN technology. This topology helped reduce hardware cost because there was no need to use one switch per department. It also kept the network organized and easier to manage.



4.2 VLAN Segmentation and Sub-interfaces

VLAN technology was one of the most important parts of the project. VLANs allowed us to separate departments logically even when multiple departments are connected to the same physical switch.

VLAN	Department
VLAN 10	Administrative Office
VLAN 20	Reception Area
VLAN 30	Doctors Workstations
VLAN 40	Medical Lab

Without VLANs, the clinic would either need more switches, or all departments would exist inside one broadcast domain which would reduce organization and security. Sub-interfaces were configured on the router interfaces so as trunk ports on the switches' interfaces to allow communication between VLANs. Instead of assigning one cable for every VLAN, multiple VLANs were carried through one physical connection per floor.

For Floor 1:

G0/0.10 was assigned to VLAN 10

G0/0.20 was assigned to VLAN 20

For Floor 2:

G0/1.30 was assigned to VLAN 30

G0/1.40 was assigned to VLAN 40

This reduced the number of physical cables while still maintaining proper VLAN separation. It was also possible to use only one single connection for all VLANs. However, we intentionally used two physical links because we wanted the floors to remain independent. If one switch or floor connection fails, the other floor can still continue operating.

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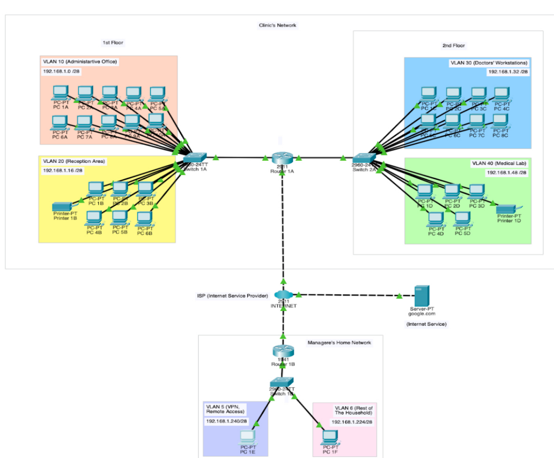
4.3 Trunk Links and IEEE 802.1Q

The connections between the router and switches were configured as trunk links. Trunk links allow multiple VLANs to pass through the same cable. Without trunking, every VLAN would require its own separate physical connection. IEEE 802.1Q was used on the trunk links. This protocol adds VLAN tags to frames so devices can identify which VLAN each frame belongs to while traveling across the trunk connection. This configuration allowed multiple VLANs to communicate efficiently while reducing hardware and cable usage.

4.4 Remote Connectivity and VPN Design

After completing the main clinic network, we added a separate home network for the clinic manager. At the beginning of the design, communication between the home network and clinic network required multiple static routes, because the clinic contains multiple VLANs and subnets. Since every VLAN has its own subnet, several routes would normally be required so the manager's home network could communicate with every clinic subnet individually. To improve security and remote accessibility, a VPN connection was implemented between the clinic network and the manager's home network.

The VPN created a secure communication tunnel between both networks and allowed the manager to access clinic resources remotely while protecting sensitive data during transmission. This made the design more secure, more realistic, and closer to real enterprise network environments.



4.5 Home Network Security Design

Inside the manager's home network, VLANs were also used to separate the manager devices from normal family devices.

The home VLAN structure included:

VLAN	Purpose
VLAN 5	Manager Devices
VLAN 6	Family Devices

This separation prevents family devices from accessing sensitive clinic resources. Access control concepts were also added so only the manager device can communicate with the clinic network while family devices remain isolated from company resources. The home network also used the last available subnet ranges from the main addressing scheme. Although any private IP address range/subnet could be used for the home network, even if it is not unique (like 192.168.1.0 /28 used for VLAN10 in the clinic), last two subnets were assigned. This was done intentionally to keep the scheme organized and simple to comprehend.

4.6 Security and Scalability Considerations

Several security and scalability ideas were considered during the design process. The use of VLANs reduced unnecessary broadcast traffic and improved logical separation between departments. The home network separation also improved security because company devices remained isolated from household devices. Scalability was also important. The subnetting plan was designed carefully so every department still has enough available host addresses for future expansion. The use of sub-interfaces also improved scalability because new VLANs can be added later without requiring large physical network changes.

4.7 Future Network Enhancements

Several future improvements can still be added to the network later. The network could include DHCP services for automatic IP assignment instead of manual configuration. Wireless access points can also be added for mobile devices and guests.

ACLs can be expanded further to control communication between departments more strictly. Firewalls and intrusion prevention systems can also be added for stronger security. Additional backup connections can also be added in the future for redundancy and better reliability.

5. IP Addressing and Subnetting Scheme

5.1 Selected Private IP Range

The network used the private IP range: 192.168.1.0/16
This range was selected because it provides enough private address space and allows easier future expansion and is commonly used for small enterprises. In addition to that, it does not waste host IP address compared to the first two proposed private ranges. From this range, the subnet block 192.168.1.0 was used for the clinic network and divided into smaller /28 subnets.

5.2 Subnetting Design

The network was divided into four /28 subnets.

Subnet Name	Network Address	Prefix	Subnet Mask	Usable Host Range	Broadcast Address	Maximum Hosts
VLAN 10	192.168.1.0	/28	255.255.255.240	192.168.1.1 - 192.168.1.14	192.168.1.15	14 Hosts
VLAN 20	192.168.1.16	/28	255.255.255.240	192.168.1.17 - 192.168.1.30	192.168.1.31	
VLAN 30	192.168.1.32	/28	255.255.255.240	192.168.1.33 - 192.168.1.46	192.168.1.47	
VLAN 40	192.168.1.48	/28	255.255.255.240	192.168.1.49 - 192.168.1.62	192.168.1.63	

This subnetting structure reduced IP waste while still providing enough addresses for future expansion.

5.3 VLAN and Gateway Assignment

Each VLAN was assigned a dedicated gateway through router sub-interfaces. The router sub-interfaces handled communication between VLANs and allowed devices from different departments to communicate correctly while still remaining logically separated. The switch management interfaces were also assigned IP addresses so the switches can be managed remotely.

5.4 Growth Planning

The clinic required support for future expansion. Every subnet was designed to support at least 20% growth. For example, the administration department currently contains 10 devices while the /28 subnet supports 14 usable hosts. This leaves enough space for future additions without redesigning the network. The same idea was applied to the remaining departments.

6. Addressing Tables

Note that first usable host IP addresses are assigned to the VLANs while the router used the last usable host address. Hosts (PCs and printers) used addresses between the first and last usable addresses.

6.1 Clinic's Network Devices Addressing Table

<table>
<tr><th>Device</th><th colspan="2">Interface</th><th>IP Address</th><th>Subnet Mask</th><th>Default Gateway</th></tr>
<tr><td rowspan="5">Router 1A</td><td rowspan="2">G0/0</td><td>G0/0.10</td><td>192.168.1.14</td><td rowspan="4">255.255.255.240</td><td rowspan="5">-</td></tr>
<tr><td>G0/0.20</td><td>192.168.1.30</td></tr>
<tr><td rowspan="2">G0/1</td><td>G0/1.30</td><td>192.168.1.46</td></tr>
<tr><td>G0/1.40</td><td>192.168.1.62</td></tr>
<tr><td colspan="2">G0/2</td><td>200.1.1.1</td><td>255.255.255.252</td></tr>
<tr><td rowspan="2">Switch 1A</td><td colspan="2">VLAN 10</td><td>192.168.1.1</td><td rowspan="4">255.255.255.240</td><td>192.168.1.14</td></tr>
<tr><td colspan="2">VLAN 20</td><td>192.168.1.17</td><td>192.168.1.30</td></tr>
<tr><td rowspan="2">Switch 2A</td><td colspan="2">VLAN 30</td><td>192.168.1.33</td><td>192.168.1.46</td></tr>
<tr><td colspan="2">VLAN 40</td><td>192.168.1.49</td><td>192.168.1.62</td></tr>
</table>

6.2 Clinic's VLAN 10 Addressing Table

<table>
<tr><th>Devices</th><th>Interface</th><th>IP Address</th><th>Subnet Mask</th><th>Default Gateway</th></tr>
<tr><td>PC 1A</td><td rowspan="10">NIC
(FastEthernet0)</td><td>192.168.1.2</td><td rowspan="10">255.255.255.240</td><td rowspan="10">192.168.1.14</td></tr>
<tr><td>PC 2A</td><td>192.168.1.3</td></tr>
<tr><td>PC 3A</td><td>192.168.1.4</td></tr>
<tr><td>PC 4A</td><td>192.168.1.5</td></tr>
<tr><td>PC 5A</td><td>192.168.1.6</td></tr>
<tr><td>PC 6A</td><td>192.168.1.7</td></tr>
<tr><td>PC 7A</td><td>192.168.1.8</td></tr>
<tr><td>PC 8A</td><td>192.168.1.9</td></tr>
<tr><td>PC 9A</td><td>192.168.1.10</td></tr>
<tr><td>PC 10A</td><td>192.168.1.11</td></tr>
</table>

6.3 Clinic's VLAN 20 Addressing Table

Devices	Interface	IP Address	Subnet Mask	Default Gateway
PC 1B	NIC (FastEthernet0)	192.168.1.18	255.255.255.240	192.168.1.30
PC 2B		192.168.1.19		
PC 3B		192.168.1.20		
PC 4B		192.168.1.21		
PC 5B		192.168.1.22		
PC 6B		192.168.1.23		
Printer 1B		192.168.1.24		

6.4 Clinic's VLAN 30 Addressing Table

Devices	Interface	IP Address	Subnet Mask	Default Gateway
PC 1C	NIC (FastEthernet0)	192.168.1.34	255.255.255.240	192.168.1.46
PC 2C		192.168.1.35		
PC 3C		192.168.1.36		
PC 4C		192.168.1.37		
PC 5C		192.168.1.38		
PC 6C		192.168.1.39		
PC 7C		192.168.1.40		
PC 8C		192.168.1.41		

6.5 Clinic's VLAN 40 Addressing Table

Devices	Interface	IP Address	Subnet Mask	Default Gateway
PC 1D	NIC (FastEthernet0)	192.168.1.50	255.255.255.240	192.168.1.62
PC 2D		192.168.1.51		
PC 3D		192.168.1.52		
PC 4D		192.168.1.53		
PC 5D		192.168.1.54		
Printer 1D		192.168.1.55		

6.6 Clinic's Cloud (Internet-Router) Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Internet-Router	G0/0	200.1.1.2	255.255.255.252	-
	G0/1	200.1.1.5		
	G0/2	8.8.8.1	255.255.255.0	
google.com Server	NIC (FastEthernet0)	8.8.8.8	255.255.255.0	8.8.8.1

6.7Home Network Addressing Table

<u>Home Network Devices Addressing Table:</u>

<table>
<tr><th>Device</th><th colspan="2">Interface</th><th>IP Address</th><th>Subnet Mask</th><th>Default Gateway</th></tr>
<tr><td rowspan="3">Router 1B</td><td rowspan="2">G0/0</td><td>G0/0.5</td><td>192.168.1.253</td><td rowspan="2">255.255.255.240</td><td rowspan="3">-</td></tr>
<tr><td>G0/0.6</td><td>192.168.1.238</td></tr>
<tr><td colspan="2">G0/1</td><td>200.1.1.6</td><td>255.255.255.252</td></tr>
<tr><td rowspan="2">Switch 1B</td><td colspan="2">VLAN 5</td><td>192.168.1.240</td><td rowspan="2">255.255.255.240</td><td>192.168.1.253</td></tr>
<tr><td colspan="2">VLAN 6</td><td>192.168.1.224</td><td>192.168.1.238</td></tr>
</table>

<u>Home's VLAN 5 and 6 Host Devices Addressing Table:</u>

Devices	Interface	IP Address	Subnet Mask	Default Gateway
PC 1E	NIC (FastEthernet0)	192.168.1.242	255.255.255.240	192.168.1.253
PC 2F		192.168.1.226		192.168.1.238

7. Device Configuration

7.1 Router Configuration

The router configuration included hostname setup, console password, enable secret password, MOTD banner, sub-interface configuration, encapsulation settings, IP address assignments, and routing configuration. The router was responsible for communication between all VLANs and between the clinic and manager home network.

Router 1A Configuration Code:

Hostname, console password, enable secret password, and MOTD banner setup:

```
# hostname Router1A
# line console 0
# console password R1lock
# login
# enable secret conflocked
# banner motd “This is The First Router in The Clinic, Authorized Access Only.”
```

Interface G0/0 and its sub-interfaces setup:

```
# int g0/0
# no shutdown
# int g0/0.10
# encapsulation dot1q 10
# ip address 192.168.1.14 255.255.255.240
# int g0/0.20
# encapsulation dot1q 20
# ip address 192.168.1.30 255.255.255.240
```

Interface G0/1 and its sub-interfaces setup:

```
# int g0/1
# no shutdown
# int g0/1.30
# encapsulation dot1q 30
# ip address 192.168.1.146 255.255.255.240
# int g0/1.40
# encapsulation dot1q 40
# ip address 192.168.1.62 255.255.255.240
```

Interface G0/2 setup:

```
# int g0/2
# ip address 200.1.1.1 255.255.255.252
# no shutdown
```

default IP route setup:

```
# ip route 0.0.0.0  0.0.0.0  200.1.1.2
> copy running-config startup-config
```

Router 1B Configuration Code:

Hostname, console password, enable secret password, and MOTD banner setup:

```
# hostname Router1B
# line console 0
# console password R1lock
# login
# enable secret conflocked
# banner motd “This is The First Home Router.”
```

Interface G0/0 and its sub-interfaces setup:

```
# int g0/0
# no shutdown
# int g0/0.5
# encapsulation dot1q 5
# ip address 192.168.1.253 255.255.255.240
# int g0/0.6
# encapsulation dot1q 6
# ip address 192.168.1.238 255.255.255.240
```

Interface G0/1 setup:

```
# int g0/1
# ip address 200.1.1.6 255.255.255.252
#no shutdown
```

default IP route setup:

```
# ip route 0.0.0.0  0.0.0.0  200.1.1.5
> copy running-config startup-config
```

7.2 Switch 1A Configuration

Switch 1A was configured for Floor 1 and included hostname setup, console password, enable secret password, MOTD banner, creating VLANs, and access port assignment.

Switch 1A Configuration Code:

Hostname, console password, enable secret password, and MOTD banner setup:

```
# hostname 1st-Floor-Switch
# line console 0
# console password 1stFSlock
# login
# enable secret conflocked
# banner motd "This is The First-Floor Switch, Authorized Access Only."
```

VLANs and access port setup:

```
# int range f0/1-10
# switchport mode access
# switchport access vlan10
# int range f0/11-17
# switchport mode access
# switchport access vlan20
# vlan10
# name vlan10-Administrative-Office
# vlan20
# name vlan20-Reception-Area
> copy running-config startup-config
```

7.3 Switch 2A Configuration

Switch 2A was configured for Floor 2 and included hostname setup, console password, enable secret password, MOTD banner, creating VLANs, and access port assignment.

Switch 2A Configuration Code:

Hostname, console password, enable secret password, and MOTD banner setup:

```
# hostname 2nd-Floor-Switch
# line console 0
# console password 2ndFSlock
# login
# enable secret conflocked
# banner motd "This is The Second-Floor Switch, Authorized Access Only."
```

VLANs and access port setup:

```
# int range f0/1-8
# switchport mode access
# switchport access vlan30
# int range f0/9-14
# switchport mode access
# switchport access vlan40
# vlan30
# name vlan30-Doctors'-Workstations
# vlan40
# name vlan40-Medical-Lab
> copy running-config startup-config
```

7.4 Switch 1B Configuration

Switch 1B was configured for the manager's house and included hostname setup, console password, enable secret password, MOTD banner, creating VLANs, and access port assignment.

Switch 1B Configuration Code:

Hostname, console password, enable secret password, and MOTD banner setup:

```
# hostname Switch1B
# line console 0
# console password S1Block
# login
# enable secret conflocked
# banner motd "This is The First Home Switch."
```

VLANs and access port setup:

```
# int f0/1
# switchport mode access
# switchport access vlan5
# int range f0/2-24
# switchport mode access
# switchport access vlan6
# vlan5
# name vlan5-VPN-Remote-Access
# vlan6
# name vlan6-Family-Network
> copy running-config startup-config
```

7.5 Trunk Configuration

Trunk links were configured between the router and switches so multiple VLANs can pass through the same connection. It also included management IP assignment.

Trunk Configuration Code on Switch 1A:

Trunk and management IP setup for interface G0/1:

```
# int g0/1
# switchport mode trunk
# switchport trunk allowed vlan 10,20
# no shutdown
# int vlan10
# ip address 192.168.1.1 255.255.255.240
# no shutdown
# exit
# ip default-gateway 192.168.1.14
# end
# int vlan20
# ip address 192.168.1.17 255.255.255.240
# no shutdown
# exit
# ip default-gateway 192.168.1.30
# end
> copy running-config startup-config
```

Trunk Configuration Code on Switch 2A:

Trunk and management IP setup for interface G0/1:

```
# int g0/1
# switchport mode trunk
# switchport trunk allowed vlan 30,40
# no shutdown
# int vlan30
# ip address 192.168.1.33 255.255.255.240
# no shutdown
# exit
# ip default-gateway 192.168.1.46
# end
# int vlan40
# ip address 192.168.1.49 255.255.255.240
# no shutdown
# exit
# ip default-gateway 192.168.1.62
# end
> copy running-config startup-config
```

Trunk Configuration Code on Switch 1B:

Trunk and management IP setup for interface G0/1:

```
# int g0/1
# switchport mode trunk
# switchport trunk allowed vlan 5,6
# no shutdown
# int vlan5
# ip address 192.168.1.241 255.255.255.240
# no shutdown
# exit
# ip default-gateway 192.168.1.253
# end
# int vlan6
# ip address 192.168.1.225 255.255.255.240
# no shutdown
# exit
# ip default-gateway 192.168.1.238
# end
> copy running-config startup-config
```

7.6 VPN Configuration

VPN was configured to secure routing and network traffic between the clinic network and the manager's home network. Configuration included installing the security package on both routers, ISAKMP protocol, shared key, transform-set, ACL, crypto map, and applying the crypto map to the routers' interfaces.

VPN Configuration Code on Router 1A:

Installing security package:

```
# license boot module c2900 technology-package securityk9
> reload
```

ACL for VPN:

```
# ip access-list extended VPN-TRAFFIC
# permit ip 192.168.1.0 0.0.0.15 192.168.1.240 0.0.0.15
# permit ip 192.168.1.16 0.0.0.15 192.168.1.240 0.0.0.15
# permit ip 192.168.1.32 0.0.0.15 192.168.1.240 0.0.0.15
# permit ip 192.168.1.48 0.0.0.15 192.168.1.240 0.0.0.15
```

ISAKMP protocol, shared key and transform-set setup:

```
# crypto isakmp policy 10
# encryption aes
# hash sha
# authentication pre-share
# group 2
# crypto isakmp key VPNKEY address 200.1.1.6
# crypto ipsec transform-set tSET esp-aes esp-sha-hmac
```

Create and apply crypto map:

```
# crypto map VPN-MAP 10 ipsec-isakmp
# set peer 200.1.1.6
# set transform-set tSET
# match address VPN-TRAFFIC
# int g0/2
# crypto map VPN-MAP
> copy running-config startup-config
```

VPN Configuration Code on Router 1B:

Installing security package:

```
# license boot module c1900 technology-package securityk9
> reload
```

ACL for VPN:

```
# ip access-list extended VPN-TRAFFIC
# permit ip 192.168.1.240 0.0.0.15 192.168.1.0 0.0.0.15
# permit ip 192.168.1.240 0.0.0.15 192.168.1.16 0.0.0.15
# permit ip 192.168.1.240 0.0.0.15 192.168.1.32 0.0.0.15
# permit ip 192.168.1.240 0.0.0.15 192.168.1.48 0.0.0.15
```

ISAKMP protocol, shared key and transform-set setup:

```
# crypto isakmp policy 10
# encryption aes
# hash sha
# authentication pre-share
# group 2
# crypto isakmp key VPNKEY address 200.1.1.1
# crypto ipsec transform-set tSET esp-aes esp-sha-hmac
```

Create and apply crypto map:

```
# crypto map VPN-MAP 10 ipsec-isakmp
# set peer 200.1.1.1
# set transform-set tSET
# match address VPN-TRAFFIC
# int g0/1
# crypto map VPN-MAP
> copy running-config startup-config
```

7.7 ACL Configuration

ACL concepts were added to allow only the manager device to communicate with the clinic network while blocking family devices.

ACL Configuration Code on Home Router 1B:

ACL setup:

```
# ip access-list extended Block-vlan6
# deny ip 192.168.1.224 0.0.0.15 192.168.1.0 0.0.0.15
# deny ip 192.168.1.224 0.0.0.15 192.168.1.16 0.0.0.15
# deny ip 192.168.1.224 0.0.0.15 192.168.1.32 0.0.0.15
# deny ip 192.168.1.224 0.0.0.15 192.168.1.48 0.0.0.15
# deny ip 192.168.1.224 0.0.0.15 192.168.1.240 0.0.0.15
# permit ip 192.168.1.224 0.0.0.15 any
> copy running-config startup-config
```

8. Connectivity Verification

Connectivity tests were performed between host devices, VLANs, and the home network. Successful ping tests verified that:

- ✓ Devices can reach their gateways.
- ✓ VLAN communication works correctly.
- ✓ Floor 1 and Floor 2 communicate successfully.
- ✓ Switch management interfaces are reachable.
- ✓ The manager's home network can communicate with the clinic network correctly.

Screenshots:

All screenshots are included in the following Google Drive link:

- ➢ Click here (Google Drive)

9. Discussion and Design Justification

The final design successfully balanced organization, security, scalability, and hardware cost. Using VLANs allowed logical separation without requiring extra switches. Sub-interfaces also reduced the number of physical cables while still allowing multiple VLANs to communicate. Using two separate physical connections instead of one single connection improved reliability because each floor can continue working independently if one switch or floor connection fails.

Another important improvement was implementing a VPN connection between the clinic network and the manager's home network. The VPN allowed secure remote access while protecting clinic data during communication between both networks. This made the design more realistic and closer to real enterprise environments where secure remote access is commonly required. The home network VLAN separation also improved security because only the manager device could communicate with the clinic network while family devices remained isolated.

10. Advanced Questions

Q1: Growth Planning Six months from now, the clinic opens a physiotherapy room on Floor 2 with 15 new computers. Does your current subnetting plan accommodate this without re addressing? Walk through the numbers and show whether it fits or what would need to change.

A1: If the clinic adds a new department with 15 devices, the current /28 subnet may not be enough because it only supports 14 usable hosts. In this case a larger subnet like /27 may be required. An efficient way to setup this new department, is to implement VLSMs (Variable length Subnet Masks) where it is assigned an IP address with the subnet mask /27. This eliminates the need to give all departments the same subnet mask.

<u>Q2: Design Trade-off</u> You chose a specific subnet size (prefix length) for each department. What would happen if you had chosen a larger subnet for every department (e.g., giving everyone a /24 regardless of size)? What is the trade-off between simplicity and efficiency?

<u>A2:</u> Using large subnet ranges for every department would make the design simpler but it would waste many IP addresses and increase unnecessary broadcast traffic. Using /28 subnets is more organized and efficient.

<u>Q3: Real-World Constraint</u> The clinic manager calls and says: *"Actually, I only budgeted for one router and two switches. Can you redesign?"* How would you respond? Could your current design be simplified to meet this constraint? What would be lost?

<u>A3:</u> The current design already uses one router and two switches. Without VLAN technology, more switches would probably be needed to physically separate departments.

<u>Q4: Troubleshooting Scenario</u> A doctor on Floor 2 reports they cannot ping the admin office on Floor 1. List at least **four specific things** you would check, in order, and explain why each check matters.

<u>A4:</u> If one department cannot communicate with another department, the first things to check would be the IP address configuration, default gateway, VLAN assignment, trunk links, sub-interfaces, and physical cable connections.

<u>Q5: Open Design Challenge</u> *(Bonus)* The clinic wants to eventually allow the manager to access the network remotely from home. Without implementing this (it's beyond this course), describe in plain language what additional device or configuration would be needed and where it would sit in your topology.

A5: To grant the manager secure and private remote access to the clinic's network, a VPN (Virtual Private Network) is needed. This creates an encrypted connection between the clinic's network and the manager's house, so sensitive information is safe from malicious attacks. The clinic router and the manager's home router are both configured for the VPN. At home, the network is split into two separate VLANs, VLAN 5 for the manager's work devices and VLAN 6 for the rest of the household. That household VLAN is denied from accessing the work devices in VLAN 5 or the clinic network, but it can still access the internet and use its services like Google (ping google.com from PC 1F in VLAN6 to check yourself). This helps keep clinic confidential info out of the family's reach. The setup is simple, and no extra devices are required in the manager's house because a router and a switch are found in every modern residence.

11. Conclusion

In this project we successfully designed and implemented a complete enterprise-style network for NorahMedical Clinic. The network achieved logical separation between departments using VLANs while still allowing communication through router sub-interfaces. Sub-interfaces reduced cable usage and improved scalability, while VPN secured routing between the clinic network and the manager's home network. The final design provided organized addressing, secure communication, future scalability, improved reliability, and realistic enterprise networking concepts while still keeping the network simple and cost efficient.